Journal of Applied Physics

Giant enhancement of nonlinear absorption in graphene oxide—Sb_2Se_3 nanowire heterostructure

Cite as: J. Appl. Phys. **125**, 025702 (2019); https://doi.org/10.1063/1.5053721
Submitted: 25 August 2018 . Accepted: 16 December 2018 . Published Online: 09 January 2019

Rajesh Kumar Yadav, J. Aneesh, Rituraj Sharma, M. Salvi, J. Jayabalan, H. Jain, and K. V. Adarsh

View Online Export Citation CrossMark

ARTICLES YOU MAY BE INTERESTED IN

Multilayer phosphorene quantum dots in an electric field: Energy levels and optical absorption
Journal of Applied Physics **124**, 124303 (2018); https://doi.org/10.1063/1.5048697

A curious observation of Pauli-Blocking in MoS_2-quantum dots/graphene hybrid system
Journal of Applied Physics **124**, 124501 (2018); https://doi.org/10.1063/1.5042278

Double carrier transport in electron-doped region in black phosphorus FET
Applied Physics Letters **113**, 193101 (2018); https://doi.org/10.1063/1.5048233

© 2018 Author(s).

Journal of Applied Physics | ARTICLE | scitation.org/journal/jap

Giant enhancement of nonlinear absorption in graphene oxide–Sb_2Se_3 nanowire heterostructure

Cite as: J. Appl. Phys. **125**, 025702 (2019); doi: 10.1063/1.5053721
Submitted: 25 August 2018 · Accepted: 16 December 2018 ·
Published Online: 9 January 2019

View Online Export Citation CrossMark

Rajesh Kumar Yadav,[1] J. Aneesh,[1] Rituraj Sharma,[1] M. Salvi,[1] J. Jayabalan,[2,3] H. Jain,[4] and K. V. Adarsh[1,a)]

AFFILIATIONS

[1]Department of Physics, Indian Institute of Science Education and Research, Bhopal 462066, India
[2]Ultrafast Spectroscopy Laboratory, Laser Physics Applications Section, Raja Ramanna Centre for Advanced Technology, Indore 452013, India
[3]Homi Bhabha National Institute, Training School Complex, Anushakti Nagar, Mumbai 400094, India
[4]Department of Materials Science and Engineering, Lehigh University, Bethlehem, Pennsylvania 18015, USA

[a)]Author to whom correspondence should be addressed: adarsh@iiserb.ac.in

ABSTRACT

Nanosheet-nanowire heterostructure provides a new platform to explore the nonlinear optical phenomena at the nanoscale. In this article, we report giant nonlinear absorption (NLA) coefficient $\approx 0.94 \times 10^3$ cm/GW at 1064 nm and 0.64×10^3 cm/GW at 532 nm in a new type of heterostructure of two-dimensional graphene oxide (GO) nanosheets and one-dimensional Sb_2Se_3 nanowires prepared by a facile hydrothermal method. The giant enhancement of NLA by the heterostructure is attributed to synergistic charge transfer between the Sb_2Se_3 nanowire and GO nanosheet through the donor-acceptor interaction. With this idea, we have developed a five-level rate equation model and numerically simulated the results. Strikingly, the heterostructure shows an excellent benchmark for the optical limiting onset of 8 mJ/cm^2 at 1064 nm and 20 mJ/cm^2 at 532 nm, that is, an order of magnitude better than graphene based materials, 2D transition metal dichalcogenides, and metal nanoparticles over a broad spectral range.

Published under license by AIP Publishing. https://doi.org/10.1063/1.5053721

INTRODUCTION

The nonlinear absorption (NLA) by 2-dimensional (2D) graphene oxide (GO) has been widely considered for applications in optical switches, optical limiters for laser safety eyewear, etc.[1,2] The optical limiting property relies on the nonlinear intensity dependence of the NLA, which offers high transmittance at low intensity while attenuates the intense laser light bursts.[1–6] Consequently, in the passive NLA based limiting, the transmission intensity is clamped at a fixed value that is determined by the NLA coefficient. Therefore, the applicability of GO in high quality optical limiting devices requires a large NLA coefficient. Besides, it must have a high dissipative optical nonlinearity, an inherently fast response time, a large dynamic range, and a broad spectral response.[1–6] Thus, the search for optical limiting and nonlinear behavior has been largely confined to chalcogenide glasses,[7] few/monolayer transition metal dichalcogenides,[8] and carbon based materials such as fullerenes, carbon nanotubes, and GO.[1–4,6,9] Among these materials, GO is much studied since the NLA can be tuned over a wide wavelength range from the visible to infrared (IR) region of the electromagnetic spectrum by controlling the oxygen functional groups.[10] However, the NLA observed in GO is comparatively weaker as compared to other materials. Many efforts have been made to improve the NLA coefficient of GO in the visible to IR region, for instance, by modifying the electronic and chemical structure through chemical functionalization and doping.[1,5,10,11] Notwithstanding, the enhancement in NLA has been minimal. In this work, we employed a nanosheet-nanowire heterostructure of Sb_2Se_3–GO for the first time to enhance the NLA. Additionally, such nanoscale heterostructures appear suitable also for an efficient platform for miniaturization and monolithic integration.[12,13]

Although it may appear complicated to grow a nanowire-nanosheet in one single structure, we present a strategy to

Published under license by AIP Publishing.

synthesize a new type of heterostructure that consists of 2D GO nanosheet and 1D Sb_2Se_3 nanowires, using hydrothermal synthesis and a dispersible template method. We have selected Sb_2Se_3 nanowires because they exhibit excellent optical, electrical, and thermoelectric properties, pressure-induced topological phase transition, etc.[14–18] Moreover, these nanowires are environmentally benign and can be prepared by cost effective wet chemistry. Furthermore, Sb_2Se_3 nanowire will act as an excellent electron donor to graphene oxide (GO). In this newly synthesized Sb_2Se_3–GO heterostructure, we report a NLA coefficient at 1064 and 532 nm. It provides a benchmark for optical limiting onset at 8 mJ/cm^2 and 20 mJ/cm^2 with the low limiting differential transmittance of 0.17 and 0.13 for 1064 nm and 532 nm laser wavelengths, respectively. We have developed a five-level rate equation model to numerically simulate the results.

EXPERIMENTAL

Synthesis of Sb_2Se_3–GO heterostructure

In our method, 0.156 g of $SbCl_3$ and 0.120 g of GO in 18 ml H_2O were mixed with 18 ml glycol and the solution was continuously stirred for 20 min. To this solution, we added 0.125 g of Se powder and 0.187 g $NaBH_4$ and stirred for another 20 min. The final solution was then transferred to a 50 ml autoclave and kept at 180 °C for 72 h. The formed Sb_2Se_3–GO heterostructure was washed several times with ethanol and distilled water. We have made GO using the modified Hummers method[19] and Sb_2Se_3 nanowire reported in Refs. 16 and 18 and the details are given in the supplemental material.

Characterization

Scanning electron microscopy (SEM) experiments were performed with Carl ZEISS (ultraplus) FE-SEM at 20 kV. PANalytical "Empyrean" powder X-ray diffractometer (PXRD) with Cu Kα radiation (1.54 Å) with a step size of 0.02° was used to record the powder X-ray diffraction patterns of the samples. "Lab Ram" high-resolution Raman spectrometer with the excitation source at 633 nm from a He-Ne laser in the confocal mode was used to record the Raman spectrum. The absorbance [$-\ln(I/I_0)$] was measured using the Cary 500 (Agilent) UV-Vis double beam spectrometer. Here, I and I_0 are the transmitted intensities through sample and reference. The open aperture Z-scan technique was used to measure the nonlinear optical absorption of different samples. Fundamental (wavelength $\lambda = 1064$ nm) and the second harmonic ($\lambda = 532$ nm) 5 ns laser pulses from a Q-switched Nd-YAG laser was used to excite the sample. The repetition rate of the laser was set at 10 Hz to provide enough time for the system to relax after each pulse. The beam was focused along the axis of a computer controlled translation stage using a 300 mm focal length lens to a diameter of $2w_0 \approx 34\,\mu$m ($\lambda = 1064$ nm) and 17 μm ($\lambda = 532$ nm) and a Rayleigh length $Z_0 \approx 3.4$ mm at $\lambda = 1064$ nm and 1.7 mm at $\lambda = 532$ nm such that it was larger than the thickness of the sample (1.0 mm liquid cell); hence, the intensity variation within the sample due to Gaussian beam propagation can be neglected. Briefly, all the data have been analyzed using a MATLAB program. We find that the goodness of fitting [coefficient of determination (R-squared)] for all evaluations is higher than 0.985 and root mean square error is less than 0.015.

RESULTS AND DISCUSSION

We strategized a hydrothermal synthesis based on the dispersible template method for fabricating Sb_2Se_3–GO heterostructure. In our method, using a low-cost hydrothermal route, Sb_2Se_3 nanowires are fabricated *in situ* on the surface of pre-synthesized GO nanosheets. A schematic representation of the sample preparation is shown in Fig. 1(a) and the details are given in the supplemental material. Figure 1(b) shows scanning electron microscopy (SEM) images of GO, which has nanosheet morphology. From Fig. 1(c), it is evident that Sb_2Se_3 have nanowires morphology. Figure 1(d) shows SEM images of an *in situ* grown Sb_2Se_3–GO heterostructure. The SEM images clearly manifest that the nanowires network is covered with GO nanosheet. Further to establish the stoichiometry of Sb_2Se_3–GO heterostructure and Sb_2Se_3 nanowires, we have performed scanning electron microscopy-energy dispersive X-ray spectroscopy (SEM-EDS) at several places. Atomic % ratio of Se and Sb in the Sb_2Se_3–GO heterostructure and Sb_2Se_3 at different places shows homogeneous chemical composition of Se/Sb = 1.5 ± 0.1, which is close to the theoretical value of 1.5 (Fig. S1).

The powder x-ray diffraction (PXRD) pattern of the Sb_2Se_3 nanowires shown in Fig. 2 is indexed to the orthorhombic (JCPDS No: 015–0861) structure and the respective lattice parameters are a = 11.62, b = 11.72, and c = 3.98 Å.[16] The intense peak at 10.68° of GO is indexed to the reflection from (200) planes that correspond to an interlayer distance of 8.27 Å. From the PXRD data, we estimate that the crystallite length of GO along the c-axis (L_c) is 84.5 Å and along the a-axis (L_a) is 28.5 Å. The PXRD pattern of Sb_2Se_3–GO heterostructure is shown in Fig. 2(a). To obtain more detailed information, we have shown magnified (020) and (120) reflections of Sb_2Se_3 in pristine and heterostructure in Fig. 2(b). A careful analysis shows that the reflection peaks a = 11.65, b = 11.78, and c = 4.03 Å shift toward lower values, indicating an increase in lattice constant of Sb_2Se_3 in the heterostructure.

Raman spectroscopy was employed to further characterize the molecular structure of the heterostructure. Shown in Fig. 3(a) is the Raman spectrum of the pristine GO, which shows characteristic D and G modes at 1332 and 1585 cm^{-1}, respectively. The edges, defects, and disordered carbon atoms contribute to the D mode, whereas the G mode originates from the graphite lattice. The peak intensity ratio (I_D/I_G) of D and G modes gives qualitative information on the degree of disorder and the graphitization degree of carbonaceous material in GO.[6,11] It can be seen from Fig. 3(a) that I_D/I_G of GO ≈ 1.10. Shown in Fig. 3(b) is the Raman spectrum of Sb_2Se_3 nanowire that contains two modes at 189 and 253 cm^{-1}. The mode at 189 cm^{-1} is due to the heteropolar Sb–Se bond stretching in the $SbSe_{3/2}$ pyramids and the mode at 253 cm^{-1}

Published under license by AIP Publishing.

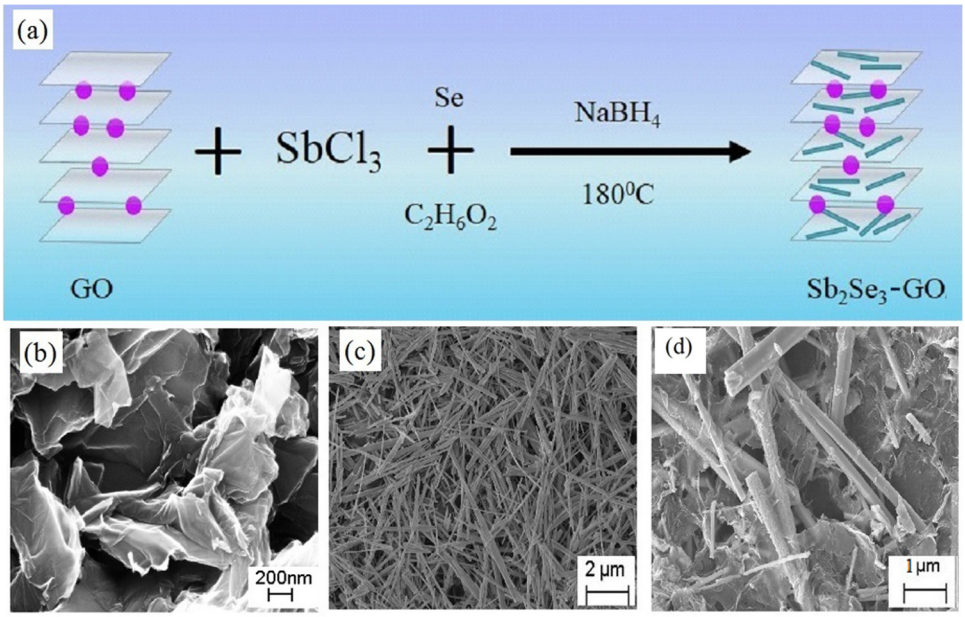

FIG. 1. (a) Schematic of the synthesis of Sb_2Se_3–GO heterostructure. SEM images of (b) GO nanosheet, (c) Sb_2Se_3 nanowires, and (d) Sb_2Se_3–GO heterostructure.

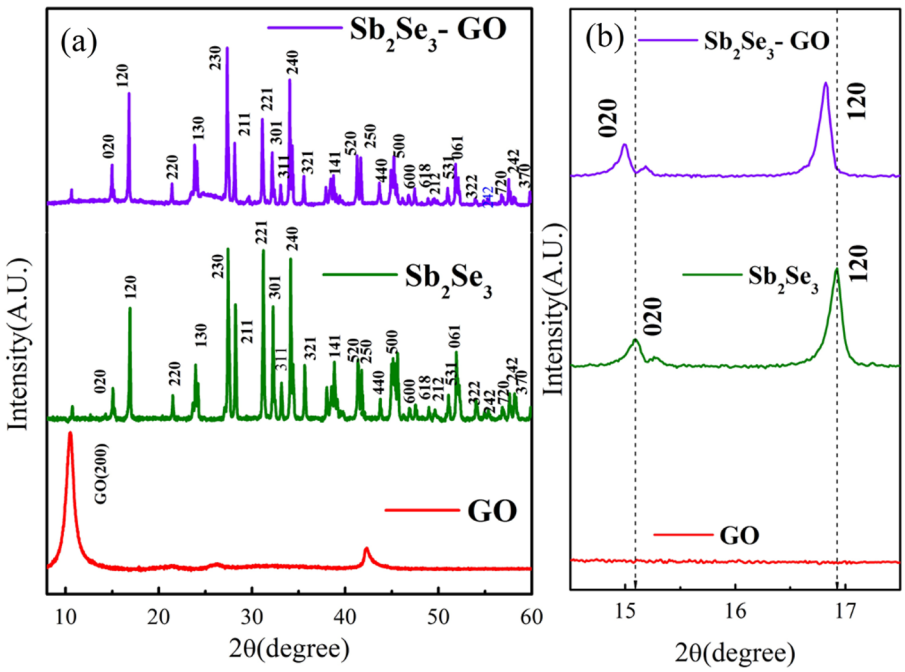

FIG. 2. PXRD spectrum of GO, Sb_2Se_3 nanowires, and Sb_2Se_3–GO heterostructure prepared by the facile hydrothermal method. (a) PXRD spectrum of GO, Sb_2Se_3 nanowires, and Sb_2Se_3–GO heterostructure. (b) Magnified (020) and (120) PXRD peaks of GO, Sb_2Se_3 nanowires, and Sb_2Se_3–GO heterostructure shown in (a). Closely looking for the other sets of (b), the 2θ positions at 15° and 17°, which correspond to the (020) and (120) reflections gradually shift toward smaller angles indicating an increase in the lattice constant of heterostructure due to strong nanosheet–nanowire interaction in Sb_2Se_3–GO heterostructure.

Published under license by AIP Publishing.

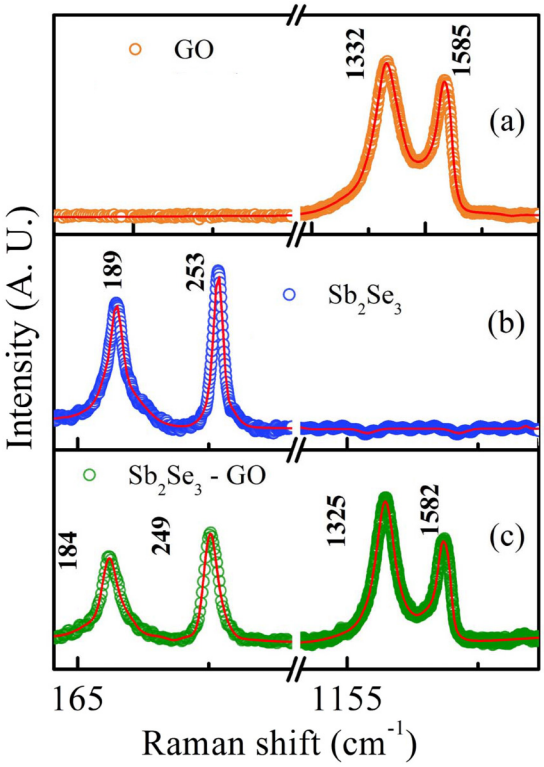

FIG. 3. (a) Raman spectrum of GO, showing two characteristic peaks within 200 cm^{-1} to 1700 cm^{-1}, which are identified as D and G modes. (b) Sb_2Se_3 nanowires, showing Sb–Se bond-stretching of the $SbSe_{3/2}$ pyramid at 189 cm^{-1} and Sb–Sb vibrations in the $Se_2Sb-SbSe_2$ configuration at 253 cm^{-1}. (c) Sb_2Se_3–GO heterostructure with four characteristic peaks within 160 cm^{-1} to 1700 cm^{-1}, giving direct evidence of successful formation of the heterostructure.

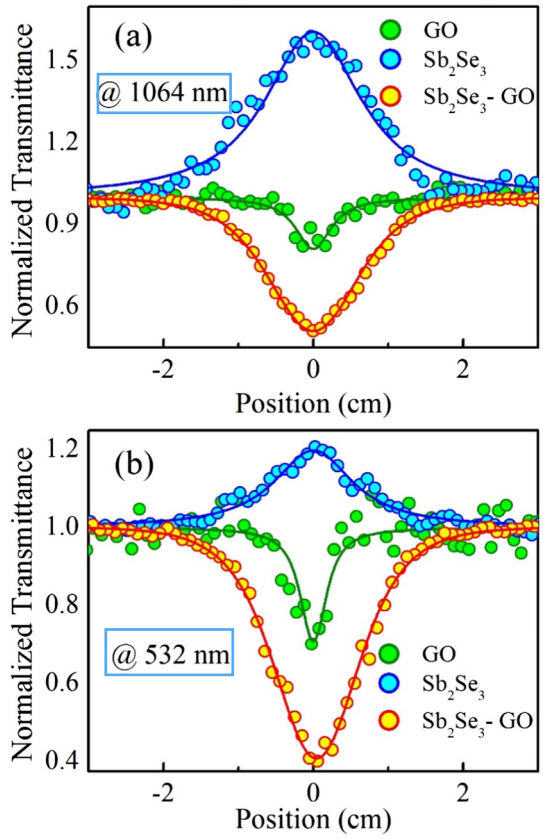

FIG. 5. Normalized transmittance as a function of position (a) at a peak intensity of 0.04 GW/cm^2 at 1064 nm pulse excitation and (b) at a peak intensity of 0.16 GW/cm^2 at 532 nm pulse excitation. The solid lines show theoretical fitting.

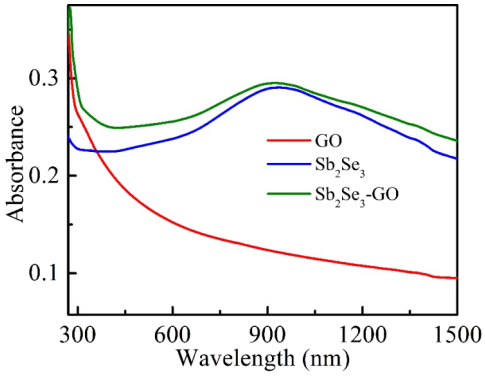

FIG. 4. Optical absorption spectrum of GO, Sb_2Se_3, and Sb_2Se_3–GO. Sb_2Se_3–GO show characteristic broad absorption extending from 300 nm to 1500 nm. It extends from IR to UV regions of the electromagnetic spectrum.

is ascribed to the homopolar Sb–Sb vibrations in the $Se_2Sb-SbSe_2$ configuration, which are well in agreement with the literature.[16,18] The Raman spectrum of the Sb_2Se_3–GO heterostructure in Fig. 3(c) can be described as a combination of D and G modes of GO and the two bond stretching modes of Sb_2Se_3. From Fig. 3, we note that the Raman spectrum of Sb_2Se_3–GO heterostructure is not merely a sum of the spectra of individual components. For instance, Raman peak positions of Sb–Se and Sb–Sb bonds are not only redshifted but are also broader. Interestingly, D and G bands of GO in the heterostructure are also redshifted to 1582 cm^{-1} and 1325 cm^{-1} together with an increase in I_D/I_G ratio. The increase in $I_D/I_G \approx 1.40$ value of the heterostructure indicates that the nucleation of Sb_2Se_3 on the surfaces of GO introduces defects in the GO network. The coexistence of the four characteristic peaks within the range 160 cm^{-1} to 1700 cm^{-1} is a direct evidence of the successful formation of Sb_2Se_3–GO heterostructure.

The optical absorption (OA) spectrum of the GO and pristine Sb_2Se_3 nanowires shown in Fig. 4 is similar to the previous results.[11,16] The direct bandgap of Sb_2Se_3 is found

Published under license by AIP Publishing.

TABLE I. NLA coefficient (β) and saturation intensity (I_s) at 1064 and 532 nm calculated by fitting the experimental data using Eqs. (S2)–(S4).

Sample	GO	Sb_2Se_3	Sb_2Se_3–GO
$\beta_{1064\ nm}$ (cm/GW)	$(0.030 \pm 0.006) \times 10^3$	...	$(0.94 \pm 0.06) \times 10^3$
$I_{s1064\ nm}$ (GW/cm^2)	...	0.019 ± 0.001	0.007 ± 0.001
$\beta_{532\ nm}$ (cm/GW)	$(0.023 \pm 0.004) \times 10^3$	...	$(0.63 \pm 0.04) \times 10^3$
$I_{s532\ nm}$ (GW/cm^2)	...	0.035 ± 0.003	0.014 ± 0.001

to be 1.12 ± 0.01 eV [Fig. S2(a)]. In contrast, GO exhibits a broad absorption spectrum with absorption peaks at 303 nm, which corresponds to the n-to-π* transition of carbonyl (C = O) group.[1,6] OA of the heterostructure shows a remarkably broad absorption over a wide wavelength range representing contributions from both Sb_2Se_3 and GO. It can be seen that the bandgap of Sb_2Se_3 in the heterostructure (1.09 ± 0.01 eV) redshifts by as much as 30 meV [Fig. S2(b)]. The large redshift in the bandgap is due to strong nanosheet-nanowire interaction in Sb_2Se_3–GO heterostructure, which was also evident from the analysis of the PXRD and Raman data. The OA spectrum of Sb_2Se_3–GO extends from near infrared (NIR) to UV regions of the electromagnetic spectrum that would facilitate its applications in optoelectronics.

To measure the NLA of Sb_2Se_3–GO heterostructure, we have used the conventional open aperture Z-scan method, which measures the total transmittance as a function of incident laser intensity.[20,21] Since the samples have strong absorption at 532 and 1064 nm, for Z-scan measurements, we have prepared the dispersions in N-Methyl-2-pyrrolidone (NMP), with a linear transmittance (T) of ≈50%. Figure 5(a) shows the Z-scan results for Sb_2Se_3–GO heterostructure, GO and Sb_2Se_3 dispersed in NMP at 1064 nm and 0.04 GW/cm^2 (peak intensity). At this moderate peak intensity, the Z-scan peak shape response of Sb_2Se_3–GO heterostructure shows a giant NLA, that is in stark contrast to the weak NLA and saturable absorption (SA) behaviors of pristine GO and Sb_2Se_3, respectively. The experimental data were theoretically fitted using nonlinear absorption equations as discussed in supplemental information to derive the NLA coefficient (β) and saturation intensity (I_s). β and I_s from the best-fit to the normalized transmittance are shown in Table I. Here, the value β of Sb_2Se_3–GO $\approx (0.94 \pm 0.06) \times 10^3$ cm/GW, which is 31 times the value of GO [$\approx (0.030 \pm 0.006) \times 10^3$ cm/GW]. To determine the role of dispersant on the enhancement of NLA of Sb_2Se_3–GO, we have shown in Fig. S3 the Z-scan traces for three different dispersants. From Table S1, it can be seen that the β values are similar when Sb_2Se_3–GO is dispersed in NMP, DMF, and ethanol. Thus, the enhancement in the NLA coefficient of Sb_2Se_3–GO when compared to GO is not due to any contribution from the dispersant.

After demonstrating the NLA of Sb_2Se_3–GO heterostructure at 1064 nm, to investigate the wavelength dependence, we have obtained the Z-scan results at 532 nm at a peak power of 0.16 GW/cm^2. Figure 5(b) shows a strong NLA for Sb_2Se_3–GO heterostructure, in contrast to the weak NLA for GO or SA of Sb_2Se_3. Therefore, we conclude that Sb_2Se_3–GO shows a giant enhancement in NLA for a broad spectral regime. From the best fit to the normalized transmittance of the Z-scan traces, we find that the β of Sb_2Se_3–GO $\approx (0.63 \pm 0.04) \times 10^3$ cm/GW, which is 27 times of GO $\approx (0.023 \pm 0.004) \times 10^3$ cm/GW. These results are tabulated in Table I. Clearly, there is an enhancement in the values of β—at least 200 times larger than the benchmark optical limiters such as carbon nanotube dispersions, and GO functionalized with C_{60} and porphyrin.[1,22,23] However, our values are slightly lower than the few- and mono-layer WS_2 for infrared excitation.[8]

Before explaining the origin of enhancement in the giant NLA of Sb_2Se_3–GO, first, we attempt to explain the nonlinear optical response of Sb_2Se_3 and GO. To explain the SA in Sb_2Se_3 nanowire, we have used a two-level model for the simplest possible case. As stated previously, 1064 and 532 nm wavelengths fall in the above-the-bandgap excitation regime of Sb_2Se_3 nanowire in the two-level model. Consequently, such excitations cause the depletion of the ground-state carriers and populate the excited state.[16] Hence, at higher intensities, the depletion of carriers in the ground state along with the filling of excited state results in the SA of Sb_2Se_3 nanowire. The NLA in GO originates from the resonant electron transfer from the ground state to excited state in the sp^3 hybridization state of oxygen functional groups.[11]

To understand the origin of the NLA in the heterostructure, let us compare the nonlinear properties of pristine GO and Sb_2Se_3 with that of Sb_2Se_3–GO heterostructure. The Z-scan trace of the GO shows a reduction in transmission when exposed to higher intensities while Sb_2Se_3 shows an increase in transmission (Fig. 5). In a heterostructure where GO and Sb_2Se_3 are independently responding to the incoming light, we should have expected a reduction in the nonlinear response of one sample by the other. Consider a range of intensity when $I < I_s$. In this range, Eq. (S3) can be approximated to

$$\alpha(I) \approx \alpha_0 + \left(\beta - \frac{\alpha_0}{I_s}\right) I\,. \tag{1}$$

Clearly, the I dependence of the overall absorption should reduce. However, this is not the case in our experiment (Fig. 5), and there is an overall increase in β of the heterostructure. A direct logical conclusion is that the carriers excited in Sb_2Se_3 through linear absorption, which led to SA, should have contributed to the NLA in the heterostructure. In the absence of any previously proposed explanation, we consider the mechanism of charge transfer that occurs between GO and Sb_2Se_3 in the heterostructure and propose the following mechanism. Recent studies have shown that GO behaves as a perfect electron acceptor.[24] Upon laser illumination, Sb_2Se_3 undergoes one-photon absorption process, which generates free electrons making it to function as an electron donor. The excited electrons in Sb_2Se_3 are transferred to the defect levels of GO and from there it absorbs another photon to go to the next higher energy level. This new charge transfer pathway (from Sb_2Se_3 excited state to GO defects state) prevents the saturation of

Published under license by AIP Publishing.

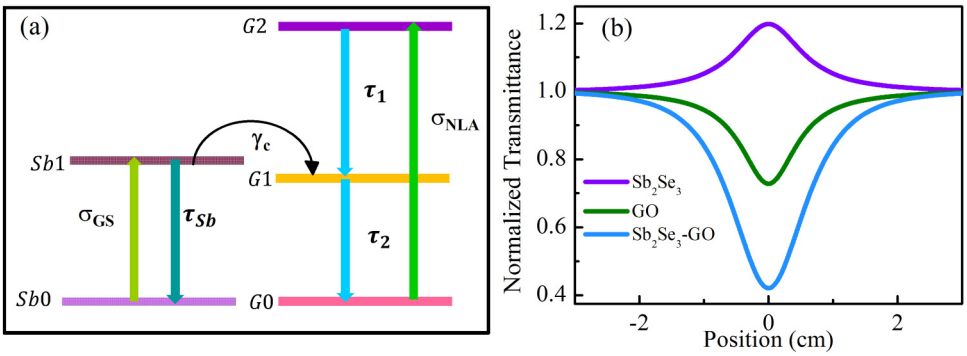

FIG. 6. (a) Five-level schematic diagram of charge transfer through donor-acceptor interaction. Sb0 and Sb1 represent ground and excited levels of the donor; similarly, G0, G1, and G2 represent ground, first excited, and second excited levels, respectively. (b) Numerically simulated data which exactly reproduced the experimental data.

Sb_2Se_3 absorption, and as a result, the net absorption of the heterostructure becomes much higher than the sum of the individual components. In the next step, we show the theoretical simulation of this process.

After demonstrating the giant enhancement of the third-order nonlinear optical response of the Sb_2Se_3–GO heterostructure, it is of great interest to explain the hallmark observation using charge transfer mediated theoretical simulation. To explain the enhancement in NLA, we have developed a five-level rate equation model based on the idea of an efficient charge transfer from the electron donor Sb_2Se_3 to the electron acceptor GO through the donor-acceptor interaction. The schematic of the charge transfer mechanism is shown in Fig. 6(a). In this model, GO behaves as a perfect electron acceptor and upon laser illumination, Sb_2Se_3 undergoes one-photon absorption process, which generates free electron making it to function as the electron donor. The excited electron of Sb_2Se_3 is transferred to the defect level of GO and from there it absorbs another photon to go to the next higher level. Such two-step absorption processes are called as the excited state absorption which is responsible for the enhancement of nonlinear absorption in the heterostructure. The population density of each level of the heterostructure can be represented by the following rate equations:

$$\frac{dN_{Sb0}}{dt} = -\frac{\sigma_{GS} I\, N_{Sb0}}{\hbar\omega_0} + \frac{N_{Sb1}}{\tau_{sb}}, \tag{2}$$

$$\frac{dN_{Sb1}}{dt} = \frac{\sigma_{GS} I\, N_{Sb0}}{\hbar\omega_0} - \gamma_c N_{Sb1} - \frac{N_{Sb1}}{\tau_{sb}}, \tag{3}$$

$$\frac{dN_{G0}}{dt} = -\frac{\sigma_{NLA} I^2\, N_{G0}}{2\hbar\omega_0} + \frac{N_{G1}}{\tau_2}, \tag{4}$$

$$\frac{dN_{G1}}{dt} = \gamma_c N_{Sb1} + \frac{N_{G2}}{\tau_1} - \frac{N_{G1}}{\tau_2}, \tag{5}$$

$$\frac{dN_{G2}}{dt} = \frac{\sigma_{NLA} I^2\, N_{G0}}{2\hbar\omega_0} - \frac{N_{G2}}{\tau_1}, \tag{6}$$

where I and γ_c are laser intensity and charge transfer rate from the donor Sb_2Se_3 nanowire to the defect state of GO, and subscripts in N (N_{Sb0}, N_{Sb1}, N_{G0}, N_{G1}, and N_{G2}) represent respective donor (N_{Sb0} and N_{Sb1}) and acceptor (N_{G0}, N_{G1}, and N_{G2}), respectively. The values of σ_{GS} and σ_{NLA} are obtained from the Z-scan experiment and can be seen in Table II. The solutions of Eqs. (2)–(6) were obtained numerically by using a Gaussian shaped pulse in both the temporal and spatial domains. The transmitted intensity through the sample is given by

$$\frac{dI(Z)}{dZ} = [-\sigma_{GS} N_{Sb0} - \sigma_{NLA} I(Z) N_{G0}]\, I(Z). \tag{7}$$

Figure 6(b) shows the numerically simulated results using Eqs. (2)–(7), which exactly reproduce the experimental data.

Having observed the giant NLA coefficient at 532 and 1064 nm, we suggest that the Sb_2Se_3–GO heterostructure can be used as a benchmark of the optical limiter, because the transmission intensity is clamped at a fixed value that is determined by the NLA coefficient. Before explaining the optical limiting application, we would like to mention the important characteristics of an ideal optical limiter. An ideal optical limiter should have high transmittance at low intensity and reduced transmittance at high intensity, a broadband response and a large dynamic range at which the device damages (irreversibly) to the limiting input. To demonstrate

TABLE II. Optical parameter for Sb_2Se_3-GO heterostructure for a wavelength λ = 1064 nm and 532 nm.

Excitation	Sample	σ_{GS} (10^{-18} cm^2)	σ_{NLA} (10^5 GM)
λ = 1064 nm	GO	0.16	0.41
	Sb_2Se_3	6.40	...
	Sb_2Se_3–GO	1.54	215.32
λ = 532 nm	GO	0.12	0.63
	Sb_2Se_3	7.63	...
	Sb_2Se_3–GO	1.33	289.46

Published under license by AIP Publishing.

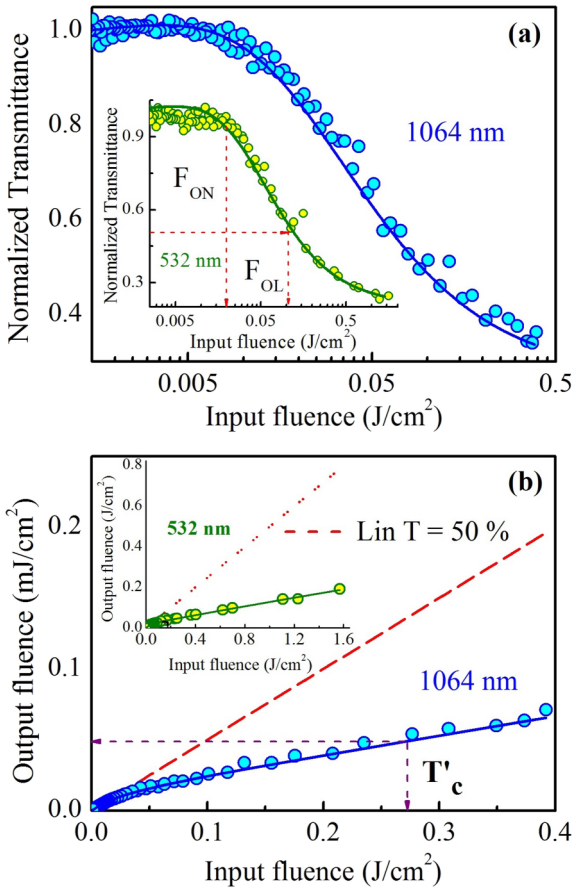

FIG. 7. (a) Plot of normalized transmittance versus input intensity of Sb_2Se_3–GO heterostructure, at 1064 nm, at peak intensity 0.08 GW/cm^2, and the inset shows the same for 532 nm at peak intensity at 0.32 GW/cm^2. (b) Output versus input intensity for Sb_2Se_3–GO heterostructure, at a peak intensity of 0.08 GW/cm^2, and the inset shows the same for 532 nm at peak intensity at 0.32 GW/cm^2. These results clearly show excellent optical limiting by Sb_2Se_3–GO from visible to IR region. The solid lines in the figure represent the theoretical fit.

the optical limiting applications of Sb_2Se_3–GO heterostructure, we have measured the transmitted fluence (F_{out}) as a function of the input fluence (F_{in}) of the laser beam. From Figs. 7(a) and 7(b), it can be clearly seen that at low input fluence, F_{out} is linearly proportional to F_{in} while following the Lambert-Beer law. As F_{in} increases, a stage reaches where F_{out} not follows the Lambert-Beer law, i.e., F_{out} is no longer linearly proportional to F_{in} and showing the deviation of the curve from the linear transmittance. In this context, it is essential to quantify the optical limiting parameters as follows: (1) onset threshold F_{ON} (the input fluence at which normalized transmittance starts to deviate from linearity), (2) optical limiting threshold F_{OL} (the input fluence at which normalized transmittance drops below 50%), and (3) limiting differential transmittance T_c' [dF_{out}/dF_{in} at high F (fluence)].[25,27]

Table III shows the values of F_{ON}, F_{OL}, and T_c' for both 1064 and 532 nm wavelengths. For 1064 nm excitation, $F_{ON} = 8$ mJ/cm^2, $F_{OL} = 85$ mJ/cm^2, and $T'_c = 0.17$ from Figs. 7(a) and 7(b). An ideal optical limiter should have very low F_{ON} and F_{OL}. The values of F_{ON} and F_{OL} for our samples are better than the standard optical limiters such as metal nanostructures, WS_2, MoS_2, carbon nanotubes, graphene nanosheet, and nanoribbons (Table SII).[25–29] We have observed a similar optical limiting performance for 532 nm [inset of Figs. 7(a) and 7(b)] and the values are $F_{ON} = 20$ mJ/cm^2, $F_{OL} = 110$ mJ/cm^2, and $T'_c = 0.13$. Thus, we conclude that Sb_2Se_3–GO is an excellent optical limiter and it defines a new benchmark for optical limiting applications.

TABLE III. Values of F_{OL}, F_{ON}, and T_C' at 1064 and 532 nm.

Sb_2Se_3–GO	1064 nm	532 nm
Lin (T)	50	50
F_{ON} (mJ/cm^2)	8	20
F_{OL} (mJ/cm^2)	85	110
T_C'	0.17	0.13

CONCLUSION

In conclusion, we have demonstrated a broadband nonlinear optical limiting behavior in the Sb_2Se_3–GO heterostructure relative to the nonlinear optical responses of pristine Sb_2Se_3 nanowires or GO nanosheet. The giant enhancement of NLA by the heterostructure is attributed to synergistic charge transfer between the Sb_2Se_3 nanowire and GO nanosheet through a donor-acceptor interaction. We have developed a five-level rate equation model and numerically simulated the results. Furthermore, Sb_2Se_3–GO heterostructure establishes a new benchmark for optical limiting at 532 and 1064 nm, which is much better than other known optical limiters such as metal nanostructures, WS_2, MoS_2, WSe_2, $MoSe_2$, carbon nanotubes, graphene nanosheet, and nanoribbons. Our results provide new insights into heterostructures for enhancing, tuning, and modulating the nonlinear optical response for potential applications in optical limiting.

SUPPLEMENTARY MATERIAL

See supplementary material for the synthesis, Z-scan theory, solvent dependent nonlinear absorption, and comparison of optical limiting property with benchmark optical limiters.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the Department of Science and Technology (Project No. EMR/2016/002520) and DAE BRNS [Sanction No. 37(3)/14/26/2016-BRNS/37245]. International Materials Institute for New Functionality in Glass at Lehigh University is acknowledged for supporting our international collaboration.

Published under license by AIP Publishing.

REFERENCES

[1]G.-K. Lim, Z.-L. Chen, J. Clark, R. G. S. Goh, W.-H. Ng, H.-W. Tan, R. H. Friend, P. K. H. Ho, and L.-L. Chua, Nat. Photonics **5**, 554 (2011).
[2]K. P. Loh, Q. Bao, G. Eda, and M. Chhowalla, Nat. Chem. **2**, 1015 (2010).
[3]M. Feng, H. Zhan, and Y. Chen, Appl. Phys. Lett. **96**, 033107 (2010).
[4]Z. Liu, Y. Wang, X. Zhang, Y. Xu, Y. Chen, and J. Tian, Appl. Phys. Lett. **94**, 021902 (2009).
[5]S. Fraser, X. Zheng, L. Qiu, D. Li, and B. Jia, Appl. Phys. Lett. **107**, 031112 (2015).
[6]R. K. Yadav, J. Aneesh, R. Sharma, P. Abhiramnath, T. K. Maji, G. J. Omar, A. K. Mishra, D. Karmakar, and K. V. Adarsh, Phys. Rev. Appl. **9**, 044043 (2018).
[7]B. J. Eggleton, B. Luther-Davies, and K. Richardson, Nat. Photonics **5**, 141 (2011).
[8]S. Zhang, N. Dong, N. McEvoy, M. O'Brien, S. Winters, N. C. Berner, C. Yim, Y. Li, X. Zhang, Z. Chen, L. Zhang, G. S. Duesberg, and J. Wang, ACS Nano **9**, 7142 (2015).
[9]P. Chantharasupawong, R. Philip, N. T. Narayanan, P. M. Sudeep, A. Mathkar, P. M. Ajayan, and J. Thomas, J. Phys. Chem. C **116**, 25955 (2012).
[10]Z.-B. Liu, X. Zhao, X.-L. Zhang, X.-Q. Yan, Y.-P. Wu, Y.-S. Chen, and J.-G. Tian, J. Phys. Chem. Lett. **2**, 1972 (2011).
[11]R. Sharma, J. Aneesh, R. K. Yadav, S. Sanda, A. R. Barik, A. K. Mishra, T. K. Maji, D. Karmakar, and K. V. Adarsh, Phys. Rev. B **93**, 155433 (2016).
[12]D. Jariwala, T. J. Marks, and M. C. Hersam, Nat. Mater. **16**, 170 (2016).
[13]C. Li, Y. Yu, M. Chi, and L. Cao, Nano Lett. **13**, 948 (2013).
[14]W. Li, X.-Y. Wei, J.-X. Zhu, C. S. Ting, and Y. Chen, Phys. Rev. B **89**, 035101 (2014).
[15]J. Anversa, S. Chakraborty, P. Piquini, and R. Ahuja, Appl. Phys. Lett. **108**, 212601 (2016).
[16]R. K. Yadav, R. Sharma, J. Aneesh, P. Abhiramnath, and K. V. Adarsh, Opt. Lett. **41**, 2049 (2016).
[17]A. Bera, K. Pal, D. V. S. Muthu, S. Sen, P. Guptasarma, U. V. Waghmare, and A. K. Sood, Phys. Rev. Lett. **110** (2013).
[18]T. Zhai, M. Ye, L. Li, X. Fang, M. Liao, Y. Li, Y. Koide, Y. Bando, and D. Golberg, Adv. Mater. **22**, 4530 (2010).
[19]D. C. Marcano, D. V. Kosynkin, J. M. Berlin, A. Sinitskii, Z. Sun, A. Slesarev, L. B. Alemany, W. Lu, and J. M. Tour, ACS Nano **4**, 4806 (2010).
[20]M. Sheik-Bahae, A. A. Said, T.-H. Wei, D. J. Hagan, and E. W. Van Stryland, IEEE J. Quantum Electron. **26**, 760 (1990).
[21]A. K. Rana, J. Aneesh, Y. Kumar, M. S. Arjunan, K. V. Adarsh, S. Sen, and P. M. Shirage, Appl. Phys. Lett. **107**, 231907 (2015).
[22]J. Wang and W. J. Blau, J. Phys. Chem. C **112**, 2298 (2008).
[23]S. Husaini, J. E. Slagle, J. M. Murray, S. Guha, L. P. Gonzalez, and R. G. Bedford, Appl. Phys. Lett. **102**, 191112 (2013).
[24]T. Musso, P. V. Kumar, A. S. Foster, and J. C. Grossman, ACS Nano **8**, 11432 (2014).
[25]H. Long, L. Tao, C. Y. Tang, B. Zhou, Y. Zhao, L. Zeng, S. F. Yu, S. P. Lau, Y. Chai, and Y. H. Tsang, Nanoscale **7**, 17771 (2015).
[26]L. Tao, H. Long, B. Zhou, S. F. Yu, S. P. Lau, Y. Chai, K. H. Fung, Y. H. Tsang, J. Yao, and D. Xu, Nanoscale **6**, 9713 (2014).
[27]A. Das, S. Ratha, R. K. Yadav, A. Mondal, C. S. Rout, and K. V. Adarsh, J. Appl. Phys. **122**, 013107 (2017).
[28]N. Dong, Y. Li, Y. Feng, S. Zhang, X. Zhang, C. Chang, J. Fan, L. Zhang, and J. Wang, Sci. Rep. **5**, 14646 (2015).
[29]H. Pan, W. Chen, Y. P. Feng, W. Ji, and J. Lin, Appl. Phys. Lett. **88**, 223106 (2006).

Published under license by AIP Publishing.